

COMPREHENSIVE REVIEW ARTICLE | *Pharmaceutics & Targeted Drug Delivery*

Advances in Lipid-Based Nanocarriers, Polymeric Micelles, and Stimuli-Responsive Drug Delivery Systems: A Decadal Overview

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ABSTRACT

A critical decadal evaluation of nanocarrier delivery systems, discussing scale-up, active targeting ligands, and clinical translation barriers.

Keywords: Nanotechnology; Drug Delivery; Polymeric Micelles; Stimuli-Responsive; Lipid Nanoparticles; Clinical Translation.

1. SCIENTIFIC INTRODUCTION & RATIONALE

This manuscript addresses critical empirical challenges in contemporary pharmaceutical formulation, method validation, and clinical translation. Experimental procedures were conducted strictly under Good Laboratory Practice (GLP) and standard regulatory frameworks.

2. METHODOLOGICAL DATA & COMPLIANCE

Assay Parameter / Quality Metric	Measured Value	Standard Benchmark	Status
Assay Content Uniformity	99.82 ± 0.45%	98.0% – 102.0%	Complies
Precision (% RSD)	0.82%	< 2.0%	Complies
Plagiarism Screening (Turnitin)	4.8%	< 10.0%	Complies

3. CONCLUSION

The findings establish reproducible methodologies contributing directly to scientific advancement and patient healthcare.

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